

MOHAMMAD MEHDI LOTFINEJAD

Principal Lead Data Scientist | Data Engineer | AI Engineer | Corporate Trainer

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PROFESSIONAL SUMMARY

Principal Lead Data Scientist, Data Engineer, AI Engineer, and HRDC-accredited Corporate Trainer with 15+ years of experience designing data platforms, building analytics and machine learning solutions, and delivering enterprise technical training. Specialized in data science, data engineering, AI/LLM applications, cloud data platforms, Apache Spark, Airflow, Databricks, Apache Doris, AWS, Azure Fabric, and modern RAG and agentic AI systems built with LangChain and LangGraph. Strong hands-on knowledge of DevOps, DataOps, MLOps, Kubernetes, Terraform, and observability, used to support reliable production-grade data and AI systems. Proven impact includes processing 2TB+ telco data daily, building 20+ production data pipelines, handling 50K+ legal documents per day, reducing manual workflow intervention by 60%, and training 500+ professionals across ASEAN.

KEY ACHIEVEMENTS

- › Lead agentic risk analytics and model monitoring solutions on Azure Fabric at CIMB, applying LangChain, LangGraph, and Rayfin to automate risk investigation and model performance oversight.
- › Led enterprise data and analytics solutions for clients including CIMB, Petronas, Shell, TNB, and regional business teams.
- › Processed 2TB+ telco data daily across multi-region AWS environments using Databricks, Apache Spark, Redshift, S3, Athena, and Lambda.
- › Built and operated 20+ production data pipelines and BI workflows with 99.9% availability for analytics users across 9 regional markets.
- › Architected AI legal-data infrastructure handling 10K+ daily case-analysis requests and 50K+ legal documents per day.
- › Reduced manual legal workflow intervention by 60% and cut case turnaround from 48 to 18 hours through workflow automation.
- › Trained 500+ professionals across ASEAN through HRDF-accredited programs in data science, Python, cloud analytics, and big data.

CORE SKILLS

Data Science & AI	Machine Learning, Predictive Analytics, NLP, LLM Applications, RAG Systems, Agentic AI, LangChain, LangGraph, Rayfin, Prompt Engineering
Data Engineering	Apache Spark, PySpark, Airflow, Databricks, Apache Doris, ETL/ELT, Data Warehousing, Data Pipelines
Cloud Data Platforms	Azure, Microsoft Fabric (Lakehouse, Data Factory, Semantic Models), Azure OpenAI, AWS, Redshift, S3, Athena, Lambda, SageMaker, GCP BigQuery, Snowflake, Apache Doris
BI & Analytics	Power BI, DAX, Tableau, Dashboard Design, Data Modeling, Business Analytics
MLOps / LLMops / DataOps	Agent orchestration and evaluation, AI workflow automation, model performance and drift monitoring, pipeline monitoring, CI/CD for data and AI workflows
Supporting DevOps Skills	Docker, Kubernetes, Helm, Terraform, Jenkins, GitHub Actions, Prometheus, Grafana, ELK
Programming & Databases	Python, SQL, PostgreSQL, MySQL, MS SQL Server, MongoDB, Neo4j, JavaScript/Node.js, Bash
Training & Leadership	Corporate Training, Curriculum Design, Technical Writing, Team Leadership, Stakeholder Management

PROFESSIONAL EXPERIENCE

Lead Data Scientist | Aug 2026 - Present

CIMB | Kuala Lumpur, Malaysia

Agentic AI and risk analytics on Azure Fabric within the group risk function.

- Lead the design and delivery of agentic risk analytics and model monitoring solutions on Azure and Microsoft Fabric, supporting credit, market, and operational risk stakeholders across the group.
- Architect the Fabric data foundation for risk analytics, including Lakehouse medallion layers, Data Factory pipelines, and Power BI semantic models serving as the single source for risk reporting.
- Build multi-agent workflows with LangChain, LangGraph, and Rayfin, orchestrating retrieval, tool calling, and human-in-the-loop review to automate risk investigation, documentation, and reporting tasks.
- Deliver automated model performance monitoring covering population and characteristic stability (PSI/CSI), discrimination (Gini/KS), calibration, and data drift, with threshold-based alerting to model owners.
- Establish LLMops practice for agentic systems, covering prompt versioning, evaluation harnesses, guardrails, cost and latency monitoring, and CI/CD for agent deployment on Azure.
- Partner with model validation, data governance, and technology teams to align agentic and monitoring solutions with internal model risk management standards and regulatory expectations.
- Lead and mentor a cross-functional team of data scientists and data engineers, setting technical direction and delivery standards for the agentic analytics roadmap.

Principal Lead Data Scientist & Corporate Trainer | May 2023 – August 2026

The Center of Applied Data Science (CADS) | Kuala Lumpur, Malaysia

- Lead data science, analytics, and data engineering engagements for enterprise clients including CIMB, Petronas, Shell, and TNB, managing delivery from proposal through deployment.
- Design Spark-based ETL and analytics pipelines on AWS and GCP to support enterprise reporting, data preparation, and machine learning use cases.
- Develop Power BI dashboards consumed by 40+ stakeholders across marketing, operations, and finance to inform quarterly strategic decisions.
- Architect cloud data-platform migration paths from legacy on-premises systems to AWS, reducing infrastructure costs by approximately 30%.
- Lead and mentor a team of 5 data scientists while designing enterprise training programs in data management, ML modeling, big data, and AI.

Chief Data and Knowledge Officer / AI Data Platform Lead | Feb 2024 - Feb 2025

Magna.ai (LawTech Startup) | Florida, USA (Remote)

Hands-on data, AI, and platform leadership role at an AI-driven legal case analysis startup.

- Owned the end-to-end AI legal data platform, combining data engineering, graph analytics, workflow automation, RAG-ready content processing, and production infrastructure.
- Architected AWS infrastructure using Terraform and Helm, supporting 15+ microservices across dev, staging, and production with reproducible deployments.
- Deployed Kubernetes-based API and workflow services handling 10K+ daily legal case-analysis requests with auto-scaling and rolling updates.
- Led Neo4j graph database and Redshift warehouse development, designing ingestion pipelines processing 50K+ legal documents daily.
- Automated legal case workflows, reducing manual intervention by 60% and cutting average case turnaround from 48 to 18 hours.
- Implemented data platform observability and governance using Prometheus, Grafana, ELK, IAM least privilege, encryption, and Vault-based secrets management.

Senior Data Engineer | Mar 2020 - Mar 2023

Axiata Digital Advertising (ADA) | Kuala Lumpur, Malaysia

Data analytics and AI subsidiary of Axiata Group, operating across 9 markets in South and Southeast Asia.

- Designed and operated 20+ production data pipelines using Databricks and Apache Spark, processing 2TB+ telco data daily across multi-region AWS environments.
- Managed AWS data infrastructure including Redshift, S3, Athena, and Lambda serving 15+ automated pipelines and BI dashboards with 99.9% availability.
- Implemented Terraform-based cloud provisioning and managed consistent dev, staging, and production environments for data engineering services.
- Administered Snowflake and Redshift data warehouses storing 20TB+ telco data, improving query performance by 40% through table design and distribution-key tuning.
- Developed Power BI and Tableau dashboards for marketing analytics across Facebook Ads, Google Ads, MobileWala, StartIO, and proprietary sources, used by 30+ analysts across 9 markets.
- Built CI/CD and monitoring practices for data services using Jenkins, CloudWatch, Grafana, and custom alerts for pipeline health, data quality, and resource utilization.
- Led and mentored 3 data engineers, coordinating data mart delivery with data science, analytics, and business teams.

Senior Data Scientist & Corporate Trainer | Jun 2018 - Feb 2020

The Center of Applied Data Science (CADS) | Kuala Lumpur, Malaysia

- Bootstrapped the company data engineering practice, building delivery processes and supporting enterprise analytics and ML solution design.
- Designed digital marketing analytics pipelines on AWS and GCP, integrating multi-channel data sources for client reporting and analytics.
- Developed technical proposals and delivered solutions covering data warehousing architecture, ML modeling, and big data processing for clients such as CIMB and TNB.
- Created and delivered HRDF-accredited programs on data science, Python, cloud analytics, and big data to 200+ professionals across ASEAN.

Big Data Engineering Team Lead | May 2016 - Jun 2018

Raha (Knowledge-Based Company) | Ahvaz, Iran

- Led 6 data engineers in building Hadoop-based academic data ingestion pipelines processing 100GB+ daily across a 10-node cluster.
- Developed PySpark and Apache Spark analytics pipelines and designed ETL workflows feeding downstream recommendation systems.

EARLIER CAREER

- Faculty Member & University Chancellor | Payame Noor University & Islamic Azad University, Iran | 2011 - 2018: Led academic teams, taught Computer Architecture, OOP, Data Structures, and Operating Systems, supervised graduate research, and published academic papers and 3 books.
- Senior Web Application Developer | Smarter Broadcast, California, USA (Remote) | 2008 - 2011: Built and maintained web applications for corporate clients, collaborating with UI/UX and QA teams.
- Business Analyst | Vira Co., Tehran, Iran | 2007 - 2008: Analyzed requirements, designed financial software solutions, and conducted UI/UX A/B testing.

CERTIFICATIONS

Harvard Business School - Business CORE Program (Business Analytics, Economics for Managers, Financial Accounting) | AWS Technical Essentials | AWS Practical Data Science with Amazon SageMaker | AWS Big Data on AWS | AWS Data Warehousing on AWS | HRDF Accredited Trainer (Malaysia)

EDUCATION

- MSc in Computer System Architecture | Islamic Azad University, Iran | 2007 - 2010
- BEng in Computer Hardware Engineering | Islamic Azad University, Iran | 2000 - 2004

SELECTED PUBLICATIONS & COURSE AUTHORIZING

- Course Author: Window Functions in SQL - Dataquest.
- Published 3 books: Object-Oriented Programming in C#, Project Management Fundamentals, and Essentials of Data Mining.
- Published 10+ peer-reviewed journal and conference papers in machine learning, neural networks, computer architecture, and data mining.

LANGUAGES

English - Professional Working Proficiency | Persian - Native